Global Superstore Sales Analytics Dashboard

Power BI Project Report

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Tools Used

Power BI | Power Query | DAX | Excel

Dataset: Global Superstore (2011–2014)

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1. Project Overview

The Global Superstore Sales Analytics Dashboard was developed using Microsoft Power BI to analyze sales performance, profitability, customer behavior, and product performance between 2011 and 2014.

The objective of this project is to convert raw sales data into meaningful business insights using interactive dashboards and DAX calculations. The dashboard allows users to filter data by year, market, category, customer segment, and other business dimensions.

The project helps business users identify high-performing products, profitable categories, customer purchasing patterns, and sales trends for better decision-making.

2. Business Problem Statement

The company generates thousands of sales transactions across different countries and product categories. However, analyzing this large amount of data manually is difficult.

The organization needs an interactive dashboard to:

- Monitor overall sales performance
- Identify profitable products
- Track monthly sales trends
- Analyze customer behavior
- Compare market performance
- Improve business decision-making

3. Dataset Description

Dataset Information:

Dataset Name: Global Superstore Dataset

Time Period: 2011 – 2014

Rows: Approximately 51,000+

Columns: 24+

Main Fields:

- Order ID
- Order Date
- Customer ID
- Customer Name
- Product Name
- Category
- Sub Category
- Market
- Region
- Country
- Sales
- Profit
- Quantity
- Discount
- Shipping Cost
- Ship Mode

Project KPIs:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin
- Average Sales per Customer
- Average Profit per Customer
- Average Orders per Customer
- Total Quantity Sold
- Average Sales per Product
- Average Profit per Product

4. Data Cleaning (Power Query)

Before creating dashboards, the dataset was cleaned using Power Query.

Cleaning Steps Performed:

- Removed unnecessary columns
- Checked duplicate records
- Verified missing values
- Changed data types
- Formatted date columns
- Created Calendar Table
- Renamed columns
- Standardized category names
- Improved data quality
- Loaded cleaned data into Power BI

5. Data Modeling & DAX Measures

Data Modeling:

- The project follows a Star Schema.

Fact Table:

- Superstore

Dimension Table:

- Calendar Table

  Relationship: Calendar[Date] → Superstore1[Order Date]

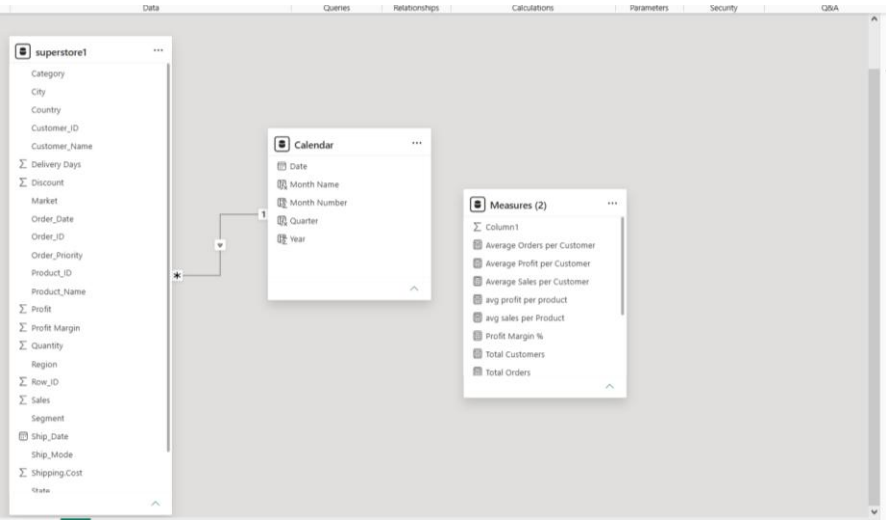


Figure: Data model showing relationship between Calendar and Superstore tables

DAX Measures Used:

1. Total Sales:

```
Total Sales = SUM(superstore1[Sales])
```

2. Total Profit:

```
Total Profit = SUM(superstore1[Profit])
```

3. Total Orders:

```
Total Orders = DISTINCTCOUNT(superstore1[Order_ID])
```

4. Total Customers:

```
Total Customers = DISTINCTCOUNT(superstore1[Customer_ID])
```

5. Profit Margin % :

```
Profit Margin % =
DIVIDE([Total Profit],[Total Sales])
```

6. Average Sales per Customer:

```
Average Sales per Customer =
DIVIDE([Total Sales],[Total Customers])
```

7. Average Profit per Customer:

```
Average Profit per Customer =
DIVIDE([Total Profit],[Total Customers])
```

8. Average Orders per Customer:

```
Average Orders per Customer =
DIVIDE([Total Orders],[Total Customers])
```

9. Average Sales per Product:

```
Average Sales per Product =
DIVIDE([Total Sales],
DISTINCTCOUNT(superstore1[Product_Name]))
```

10. Average Profit per Product:

```
Average Profit per Product =
DIVIDE([Total Profit],
DISTINCTCOUNT(superstore1[Product_Name]))
```

6. Sales Performance Dashboard



Figure: Sales Performance Dashboard – Global Superstore (2011–2014)

Dashboard Description:

This dashboard provides an overview of overall sales performance. It displays:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Monthly Sales Trend
- Sales by Category
- Top Products
- Global Sales Distribution
- Customer Segment Analysis

Key Insights:

- Total Sales reached approximately $13M.
- Technology category generated the highest sales.
- Consumer segment contributed the highest revenue.
- Sales increased significantly in the last quarter.
- Top-selling products contributed a major share of total revenue.

7. Profit Analytics Dashboard



Figure: Profit Analytics Dashboard – Global Superstore (2011–2014)

Dashboard Description:

This dashboard focuses on profitability analysis.

Key Insights:

- Total Profit exceeded $1.5M.
- Overall Profit Margin is approximately 11.6%.
- Technology category generated the highest profit.
- Monthly profit shows an upward trend.
- Central region contributed the highest profit.
- Top profitable products generated significant earnings.

8. Customer Analytics Dashboard



Figure: Customer Analytics Dashboard – Global Superstore (2011–2014)

Dashboard Description

This dashboard analyzes customer purchasing behavior.

Key Insights

- Total Customers: approximately 5K.
- Average Sales per Customer: $2.59K.
- Average Profit per Customer: approximately 301.
- Consumer segment contains the highest number of customers.
- APAC market recorded the highest customer orders.
- Top customers contribute significantly to total sales.

9. Product Analytics Dashboard

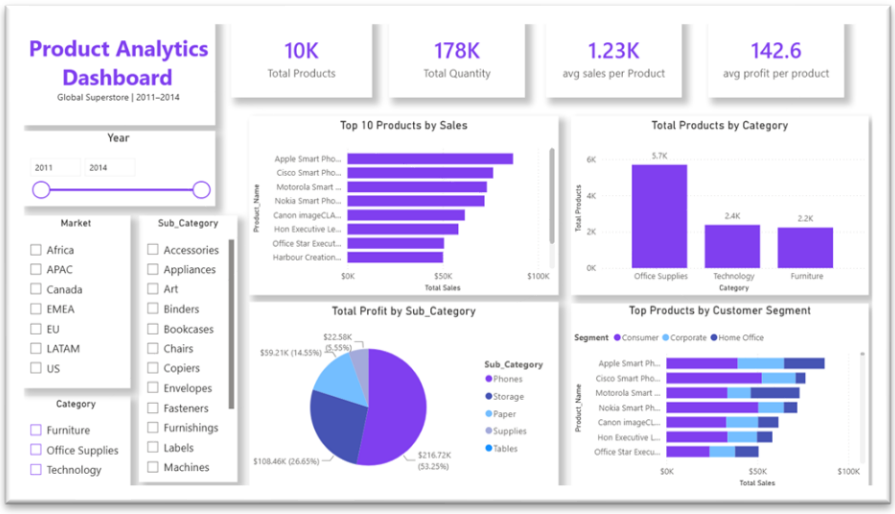


Figure: Product Analytics Dashboard – Global Superstore (2011–2014)

Dashboard Description

This dashboard evaluates product performance.

Key Insights

- Total Products: approximately 10K.
- Total Quantity Sold: approximately 178K.
- Office Supplies contains the highest number of products.
- Phones generated the highest sub-category profit.
- Top-selling products account for a large portion of revenue.
- Product performance differs across customer segments.

10. Business Insights

1. Technology is the best-performing category by sales.
2. Technology also generates the highest profit.
3. Consumer segment contributes the most revenue.
4. Office Supplies contains the highest number of products.
5. Monthly sales increase toward the end of the year.
6. Profit follows a positive trend over time.
7. Top 10 products generate significant sales.
8. Average customer spending is around $2.59K.
9. Average profit per customer is approximately $301.
10. APAC market has the highest order volume.
11. Phones are the most profitable sub-category.
12. Product performance varies by customer segment.
13. Interactive filters improve analysis.
14. Calendar table enables time-based analysis.
15. DAX measures provide dynamic KPIs.
16. Customer segmentation supports targeted marketing.
17. Regional performance differs across markets.
18. Profit margins vary by category.
19. Data modeling improves dashboard performance.
20. Interactive dashboards enable faster business decisions.

11. Recommendations

Based on the insights derived from the dashboards, the following recommendations are proposed:

- Increase investment in the Technology category, as it consistently drives the highest sales and profit.
- Introduce targeted promotions for the Consumer segment to further capitalize on its strong revenue contribution.
- Review pricing and discount strategy for low-margin categories to improve overall profitability.
- Strengthen supply and marketing focus in the APAC market given its high order volume.
- Expand the Phones sub-category product range, since it generates the highest profitability.
- Launch customer loyalty programs aimed at top customers to sustain long-term revenue.
- Use monthly sales trend data to plan inventory ahead of the year-end demand increase.

12. Challenges Faced

- Handling a large dataset (51,000+ rows) required careful attention to model performance.
- Identifying and resolving inconsistent category and sub-category naming during data cleaning.
- Establishing accurate date-based relationships between the Calendar table and the Superstore fact table.
- Designing DAX measures that remained accurate across all filter and slicer combinations.
- Balancing visual density with dashboard readability across four separate analytical pages.
- Ensuring consistent formatting and color themes across all dashboard pages.

13. Future Scope

- Integrate real-time or scheduled data refresh instead of static historical data.
- Add predictive analytics using Power BI's forecasting features to project future sales and profit.
- Incorporate additional external data such as marketing spend or economic indicators for deeper analysis.
- Develop a mobile-optimized version of the dashboard for on-the-go decision-making.
- Expand the model to include return/refund data for a more complete profitability picture.
- Automate report distribution using Power BI Service subscriptions.

14. Conclusion

- The Global Superstore Sales Analytics Dashboard successfully transforms raw transactional data into actionable business insights through interactive Power BI dashboards. By combining clean, well-modeled data with meaningful DAX measures, the project enables stakeholders to monitor sales, profit, customer, and product performance across the 2011–2014 period.
- The dashboard highlights Technology as the leading category by both sales and profit, Consumer as the strongest customer segment, and APAC as a high-volume market, providing a clear foundation for data-driven business decisions. Overall, this project demonstrates practical proficiency in Power Query, data modeling, DAX, and dashboard design.